Designing a Visual Analytics Framework for Monitoring Insect Pests Populations

Case of study: South Tyrol

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South Tyrol, on the border between Austria and northern Italy, is one of Europe's leading fresh apple producers [1].

In recent years, pest insects have threatened this production, causing severe economic losses that are only increasing with the arrival of invasive species driven by climate change [2][3].

This work develops an end-to-end visual analytics workflow for insect pest monitoring: expert requirements are translated into an integrated data model that unifies climate, phenological, and monitoring data from heterogeneous sources, enabling visual exploration to support pest populations monitoring across the region.



Fig. 1: Apple production in S.T. (The South tyrolean apple, n.d)

OBJECTIVES

RO1: Identify and formalize the visualization, analytical, and data organization requirements through expert consultation.

RO2: Design and implement a data integration schema that standardizes heterogeneous monitoring, phenological, and climate data for spatial, temporal, and thematic analysis oriented with FAIR.

RO3: Design, develop, and evaluate a map-based visual analytics prototype that visualizes this integrated data and supports interpretation of pest population dynamics.



Fig. 2: Implemented specie: H. halys. (Wikimedia, 2017)

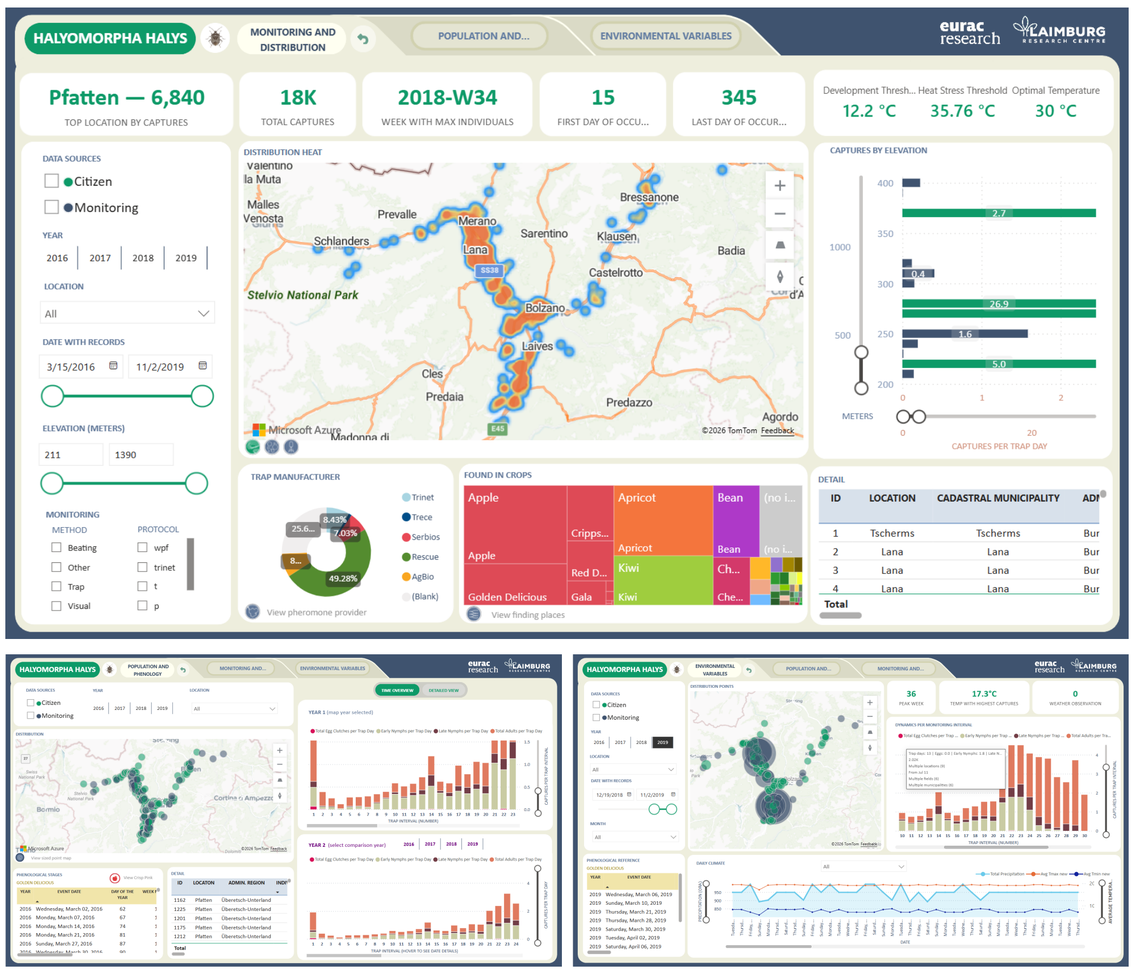


Fig. 3: Visual Analytics thematic pages (final version). Implementation in Power BI.

WORKFLOW

The methodology employs a waterfall methodology in which each step could only begin once the previous one was complete. It's composed by six phases:

1. The data was collected, preprocessed and explored in its multiple formats.
2. The user needs identification through an expert consultation and understanding data strucutures in field work.
3. First interface proposal.
4. Data integration model design.
5. Implementation in Power BI.

RESULTS

Data exploration produced an initial inventory of data fields from five monitoring species. Only Halyomorpha halys was implemented; the others served as context for dataset structures.

Alongside identifying 19 user requirements, this shaped the data integration schema, with the temporal axis as the main organizing element.

The VA was then implemented in Power BI. After three evaluation rounds, it became a refined tool with three thematic pages: distribution and monitoring, population and phenology, and environmental variables.

EVALUATION

The prototype was evaluated in three phases to ensure its usefulness.

1. An internal review with the supervisor.
2. A feedback session with the stakeholder representative.
3. A structured survey accompained with a clarification sessions.



Fig 4: Feedback session at EURAC-CSS.

CONCLUSIONS

This research offers value beyond the tool itself by highlighting important factors in the process.

The workflow guideline was shaped by user requirements (RO1), which also informed the development of the data schema through an understanding of the monitoring process (RO2). This reflects the essence of cartography: offering new ways to understand space and time, while cartographers continually learn from their users.

Future work will address how to manage the uncertainty inherent in representing insect monitoring processes.

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